

The Large and Likely Inefficiency of Stable Matching Mechanisms

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Stability can be inefficient

- School choice mechanisms: often (student-proposing) Deferred Acceptance
- DA is attractive: it selects the student-optimal stable matching
- Stability eliminates justified envy, but might be Pareto inefficient
- Main question:

How many students can be improved after DA without harming anyone?

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Main Takeaway

In large, random market: **almost every** student can be Pareto-improved in every stable outcome

A simple DA outcome

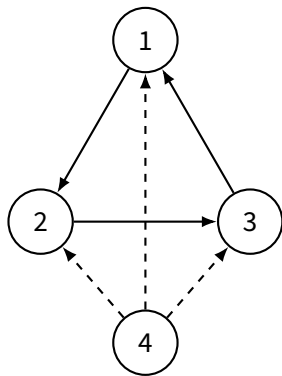
Suppose DA assigns four students to four schools:

Student	DA assignment	Schools preferred
1	a	b
2	b	c
3	c	a
4	d	a, b, c

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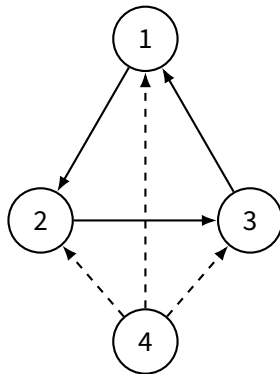
From DA to an envy graph

Definition

The DA envy graph $G_{\text{DA}}(P)$ has:

- node = student,
- edges: $i \rightarrow j$
if student i prefers student j 's DA school to her own.

A directed cycle means that the students in the cycle can trade DA seats and all strictly improve.



$$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$$

Cycles characterize improvable students

Key characterization (Erdil '14, Dur et al. '19)

A student is improvable if and only if she belongs to a directed cycle in $G_{\text{DA}}(P)$.

Trade along the cycle: Every student in cycle gets a school she prefers to her DA assignment

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Graph-theoretic translation

Studying Pareto improvements after DA is **equivalent** to studying cycles in the DA envy graph

Strongly connected components

Definition

Set of nodes is **strongly connected** if every node can reach every other node via directed paths

Strongly connected component (SCC) = *maximal* strongly connected set.

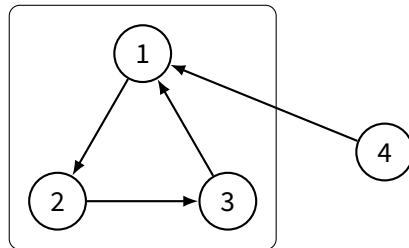
Strongly connected components

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Set of nodes is **strongly connected** if every node can reach every other node via directed paths

Strongly connected component (SCC) = *maximal* strongly connected set.

- A non-trivial SCC has at least two nodes
- Every node in a non-trivial SCC lies on some directed cycle
- $\implies$ every student in a non-trivial SCC is improvable



non-trivial SCC

Why large markets change the picture

In a large random market, DA generates many applications.

- student i applies to the school eventually assigned to student j
 $\implies i$ envies j when i is rejected there.
- Many applications create many envy edges
- A dense directed graph is unlikely to fragment into many isolated pieces

Network intuition

Large random DA envy graphs tend to develop a giant SCC.

If almost all students belong to one giant SCC: almost all students are improvable.

Formal model

A school choice problem is

$$P = (I, S, (\succ_i)_{i \in I}, (\triangleright_s)_{s \in S}).$$

- I : set of students.
- S : set of schools.
- Each student i has strict preferences $\succ_i$ over schools.
- Each school s has strict priority $\triangleright_s$ over students.
- Baseline random model:

$$|I| = |S| = n,$$

and all preferences and priorities are drawn independently and uniformly at random.

DA produces matching $\text{DA}(P)$ and induces the envy graph $G_{\text{DA}}(P) = (I, E(\text{DA}(P)))$

Main theorem

Theorem

Fix any $\varepsilon > 0$. With high probability, the DA envy graph $G_{\text{DA}}(P_n)$ has a unique giant SCC containing at least $(1 - \varepsilon)n$ students.

Consequence

The fraction of unimprovable students converges to zero in large markets.

In large random markets: almost every student can be made strictly better off than under DA without making anyone worse off

Beyond existence: simultaneous improvements

Previous theorem: almost all students are individually improvable

In the paper: these improvements are not isolated

Cycle-packing result

W/ high prob., every maximal packing of disjoint trading cycles covers at least $(1 - \varepsilon)n$ students.

- Almost all students can be improved simultaneously
- Efficient mechanisms that P-dominate DA improve essentially the same no. of students
- Policy: focus on other criteria, such as stability loss, fairness, and incentives

Stable outcomes are likely to be inefficient on a large scale

- DA is stable, but stability can produce large inefficiency
- (DA) outcomes $\leftrightarrow$ envy graphs
- Cycles characterize improvable students: SCC
- In large random markets: the DA envy graph contains a unique giant SCC
 $\implies$ Almost all students are improvable relative to DA.
- Other stable mechanisms: covered because DA is student-optimal stable one

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